

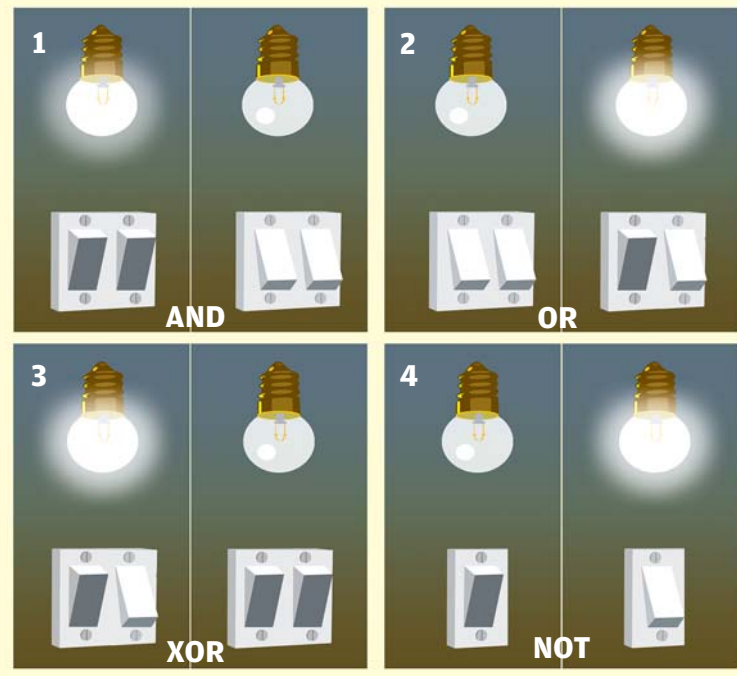
are also 1. To construct any chip, you're going to need the most basic AND, OR, NOT and XOR (eXclusive OR) gates. Today, each gate serves one specific purpose, so if you wanted a chip that could perform AND operations and OR operations, you'd need to fabricate one with both gates, even if you don't need them both at the same time. With Ditto's idea, however, you'd just need to fabricate one circuit, which can then become any gate you want it to be with only a slight programmatic adjustment.

We always talk about chips dealing with ones and zeroes, but we overlook the fact that at the bottom of it all, they're actually dealing with electric current. A typical IC today treats any voltage below approximately 0.8 volts as a logical 0, and any voltage above approximately 3.1 volts as a logical 1. Life between these two thresholds is grey area, and will generally confound the chip in question. Now, imagine if you played around with these thresholds a little.

Let's take an example: we have at our disposal an OR gate, and give it one input of 0.4 volts (a logical 0), and 5 volts (a logical 1) at the other input, resulting in an output of, say, 4 volts (still a logical 1). Now let's change the threshold for 1 to 4.5 volts. The inputs are still 0 and 1, but the output will now be taken as 0, and poof! Our OR gate is now an AND gate. The example is crude, and purists would probably come after us with torches lit, but you get where the chaos angle comes in. Since there's really no limit to how many new thresholds you can set, each change can result in the formation of a new gate. The changed outputs of these gates will affect the outputs of the next set of gates, then the gates after those, and in full compliance with the Butterfly Effect, a tiny change in the threshold (the butterfly's wing flapping) has enormous consequences (the hurricane). By manipulating thresholds in this manner, gates can be transformed on a wide scale, making it possible to create complex chips that can change their behaviour whenever we want them to. These chips will be built using *chaotic elements*—devices that have inherent chaotic behaviours, and it isn't limited to electrical circuits; we'll soon be seeing lasers and even neurons joining in. Ditto's company, ChaoLogix Inc., has already begun work on the Reconfigurable Chaotic Logic Gate Array (RCLGA), which will subsequently be interfaced with an existing operating system to show off its capabilities.

Programming The Madness

It's probably not that awe-inspiring to talk of reconfiguring one or a few gates, but think about the chaos that will ensue when you deal with circuits that contain thousands, even millions of gates—one false move and you could turn your CPU into a sound card instead of a memory chip, and you know you don't want that! Programming these chips is going to be quite difficult, and if programmers find that programming for the chaotic chip is more trouble than it's worth, we wouldn't put it past them to reject the idea altogether and carry



How Logic Gates Work: (1) The AND gate only turns on when all its inputs are on (2) The OR gate turns on when any one, or all inputs are on (3) The XOR gate is like the OR, but turns off when all inputs are on (4) The NOT gate is the exact opposite of the input—it turns off when the input is on, and vice-versa

on writing code for existing single-purpose hardware. ChaoLogix will also be developing a programming language for the RCLG arrays; time will tell how programmers react to it.

Why?

Doubtless you're trying to see what's so great about chaotic chips other than their "I'm-so-cool-I-can-do-anything" claim to fame. The most obvious advantage is, of course, the end of the single-purpose chip. If you were a chip manufacturer, and spent five years and millions of dollars on a "sure thing," only to find out that nobody in the world wanted it in the first place, you'd be feeling mighty stupid, wouldn't you? Instead of wasting years developing the innards of an otherwise silly device like, say, the next revolutionary toaster, manufacturers will be able to use a single, generic chip and program it to do serve their purpose, leaving them time and money to focus on more important things.

Designing and fabricating a chaotic chip is going to be a lot easier and cheaper than current chips, especially now that everyone's straining to get as many transistors into as small an area as possible. And really, who doesn't love cheap technology?

Finally, Ditto claims that because the chaotic chip can switch roles in a single cycle of the system's clock signal, smart design can enable it to perform many more operations per second than existing chips, resulting in faster computing than we know today. Imagine a computer with two chaotic chips—at a given time, we could have one dedicated to processing and one playing the role of system memory. If we need more number-crunching power, we could then have both playing the role of processor!